

Dagelinckx, Michel

Van: Prins HA (Harald) <harald.prins@movares.nl>
Verzonden: woensdag 19 februari 2025 17:18
Aan: Dagelinckx, Michel; Shokri, Maryam
Onderwerp: [EXTERNAL] RE: Tilburg380 voltage funnel

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Dear Michel,

I made quick & dirty soil models to check if a 520 Ohm.m top layer and a 500 Ohm.m bottom layer was realistic. I put all the measurements in one model. I did a few variations by deleting some of the very high values. That gave top layer around 110 Ohm.m and bottom layer around 60 Ohm.m with a layer thickness of around 4 meter.

Your models for the separate measurements give some different results. They deliver also some higher values for the top layer. But the bottom layer in your models is also (much) lower than 500 Ohm.m.

I also think measurement 2 is less reliable because of the high values.

All your models (except TLB380_1) (and TLB380_2 rejected) give also a lower value for the bottom layer, around 50-60 Ohm.m. Thus the bottom layer of 500 Ohm.m seems to much worst case based on all the models you and I made.

Measurement locations 1 and 2 seem to be on dryer soil with higher vales to be expected.

Measurement 3 and 4 are in wetter soil with lower values to be expected. The results for 1 and 2 (and 59N) differ form 3 and 4 (and W44H0027) as can be expected.

Met vriendelijke groet,

Prins HA (Harald)

Senior adviseur EMC en aarding

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Werkdagen: Ma t/m Do



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Van: Dagelinckx, Michel <Michel.Dagelinckx@tennet.eu>

Verzonden: woensdag 12 februari 2025 12:47

Aan: Prins HA (Harald) <harald.prins@movares.nl>; Shokri, Maryam <Maryam.Shokri@tennet.eu>

Onderwerp: RE: Tilburg380 voltage funnel

Harald,

Thanks for your report on the soil models of Tilburg 380. I have two comments/concerns:

- Your 3 models do not fit all measurements points very well. This is because the different measurement sets are very different. If you make a soil model for each dataset separately, you get a nicer fit for all four set; see my plots that I have added in a separate sheet "plots MDA 11-2-2025" to your report. Basically, one can say that the measurement (1 and 2, also our 59N measurement) on the north side of the old "Loonse Spinderspad" are more high ohmic than the measurements (3 and 4, also W44H0027) on the south side. This might be because of the water presence at "de waterberging". Now station Tilburg 380kV will be half on the north side and half on the southside, but of course the earth grid is one complete structure. So, in real, I suppose the less high ohmic south part will conduct more current to ground than the more high ohmic north part. The best way to proceed (for the contractor in a later stage; I suppose you are finished with this) might be:
 - to calculate a Ra for the south part (for half an earth grid) using the most restricted soil model TLB_3 in that area.
 - to calculate a Ra for the north part (for half an earth grid) using soil model TLB_1 (profile p1 is most restrictive, so use this) in that area; I would not use TLB_2 because it looks so unrealistic high ohmic to me.
 - Because the soil in the substation area is conditioned to be able to build there the first 1 or 2m could have changed and might be a more uniform soil now. Maybe 500 Ohm.m for clean soil is a good estimation for it.
 - Calculate the voltage funnel with the amount of SC flowing in the south side with model TLB_3; maybe top layer changed to 500 Ohm.m
 - Doing the same for the north part where the top layer might be reduced to the same 500 Ohm.m. Or check what happens if it stays 1254 Ohm.m.
- ~~Another HIGHLY IMPORTANT thing is that I see you are using a different approach when making soil models for Wenner measurements. You are using $AB/2=1,5 \times \text{pen distance}$. I agree when reading the file "Rho32.dat", this is the right way to do this. Unfortunately, I simply filled in the pen distance in the AB/2 column for all my Wenner models. This means that the H1 I calculated is 1,5 times too small; luckily Rho1 and Rho2 stay the same. This concerns 61% of all my soil models for ZW3800; the other 39% are Schlumberger models where the correction is not needed. I still have to check what this means for the results on the calculated voltage funnels. I hope not too dramatic...~~

MDA 27-5-2025: Niet relevant voor discussie TLB380

Met vriendelijke groeten / Kind regards / Mit freundlichen Grüßen,

Ir. M.H.P. (Michel) Dagelinckx

EMC Engineer

Large Projects Netherlands Engineering | ZuidWest-Oost | Standaardisatie EMC

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Van: Prins HA (Harald) <harald.prins@movares.nl>

Verzonden: dinsdag 14 januari 2025 11:02

Aan: Shokri, Maryam <Maryam.Shokri@tennet.eu>; Dagelinckx, Michel <Michel.Dagelinckx@tennet.eu>

Onderwerp: [EXTERNAL] RE: Tilburg380 voltage funnel

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Dear Maryam, Michel,

I made some soil models based on the 4 measurements (8 profiles) in the report Maryam send. There are some very high outcomes in some of the measurements, but including or excluding these values from making a soil model does not have a large effect on the soil model.

I made 3 soil models;

- All the measurement values
- Excluding measurement 2, profile 1 and 2
- Excluding all values higher than 5000 Ohm.m

The 3 soil models give similar values. I do not know how they came up with 520 and 500 Ohm.m for the layers in the soil model.

Met vriendelijke groet,

Prins HA (Harald)

Senior adviseur EMC en aarding

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Werkdagen: Ma t/m Do



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Van: Shokri, Maryam <Maryam.Shokri@tennet.eu>

Verzonden: maandag 13 januari 2025 13:20

Aan: Prins HA (Harald) <harald.prins@movares.nl>; Dagelinckx, Michel <Michel.Dagelinckx@tennet.eu>

Onderwerp: Re: Tilburg380 voltage funnel

U ontvangt niet vaak e-mail van maryam.shokri@tennet.eu. [Ontdek waarom dit belangrijk is](#)

Dear Harald,

I found the file that you mentioned in your email. I will check it. I attached the copy of the document to this email in case that you want to have a look at it.

Best regards,
Maryam

From: Prins HA (Harald) <harald.prins@movares.nl>

Sent: Monday, January 13, 2025 1:00 PM

To: Dagelinckx, Michel <Michel.Dagelinckx@tennet.eu>; Shokri, Maryam <Maryam.Shokri@tennet.eu>

Subject: [EXTERNAL] RE: Tilburg380 voltage funnel

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Dear Michel, Maryam,

The soil conditions around Tilburg are not favorable. The first soil measurements were Wenner measurements, and gave high values. See in Meridian 002.678.22 0768732, 002.678.22 0768824, 002.678.22 0768825 and 002.678.22 0768826. They were marked as not consistent. After that Marcel Janssen with LPH performed some driven rod measurements. 002.678.22 0768722 in Meridian. After that also soil measurements for the towers were performed more recent.

The soil model in the step- and touch voltage report is very worst case, 520 and 500 Ohm.m in both layers. (the 3000 Ohm.m for the thin gravel layer is ok). They reference to 002.678.00 0967730 for Wenner measurements. I do not have this Meridian reference available, so I do not know if these are the first non consistent Wenner measurements, or other Wenner measurements. Maybe you can check this reference 002.678.00 0967730 if these are the first Wenner measurements? Your soil models for the towers near TLB380 have much lower values.

The soil model used for the step- and touch voltages is to worst case, and the value of the resistance of the earthing grid is too high.

Met vriendelijke groet,

Prins HA (Harald)

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Van: Dagelinckx, Michel <Michel.Dagelinckx@tennet.eu>

Verzonden: donderdag 9 januari 2025 07:54

Aan: Shokri, Maryam <Maryam.Shokri@tennet.eu>; Prins HA (Harald) <harald.prins@movares.nl>

Onderwerp: RE: Tilburg380 voltage funnel

Harald,

This can not be true. I calculated $R_a=0,778$ for the foundation of M1205 alone. This party is using a far to worst case soil model. See table 8; screenshot below.

Tabel 8: Aardingsweerstand nemen voor berekeningen

Diepte [m]	Aardingsweerstand [$\Omega \cdot m$]	Materiaal
0 – 0,15	3000	Kiezel / grind
0,15 – 3,15	520	Zand
3,15 – ∞	500	Zand

Furthermore they inject about 30kA into the stations earth grid. This will never happen in real, because if the R_a of the station is really that high, almost no SC wants to go into the soil there. Most current will flow thru earthing wires back to the power sources feeding the SC. You can not use this data for estimating the voltage funnel of this station.

Met vriendelijke groeten / Kind regards / Mit freundlichen Grüßen,

Ir. M.H.P. (Michel) Dagelinckx

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Van: Prins, Harald <Harald.Prins@tennet.eu>

Verzonden: vrijdag 20 december 2024 10:54

Aan: Shokri, Maryam <Maryam.Shokri@tennet.eu>; Dagelinckx, Michel <Michel.Dagelinckx@tennet.eu>

Onderwerp: Tilburg380 voltage funnel

Dear Maryam, Michel,

Attached is the step and touch voltage report of Tilburg.

On page 58 the 1-fase fault current to ground is 29073 A. The GPR is 27077 V. The resistance to earth of the earthing grid must be 0,931 Ohm. That seems very high. The older requirement was maximum 0,1 Ohm for a high voltage station. This will lead to a large voltage funnel of station TLB380. On page 36 in figure 6 is only a small part of the potential distribution given. It is not visible how fast the potential decreases.

Met vriendelijke groeten / Kind regards / Mit freundlichen Grüßen,

Harald Prins

Engineer EMC en aarding
Large Projects - Area South

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